

University of Prishtina “Hasan PRISHTINA”



Faculty of Electrical and Computer Engineering
Master in Computer and Software Engineering, 2025-2026
Course: Data Preparation and Visualization

Week XI
Data Evaluation

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Metrics for Performance Evaluation

- Focus on the predictive capability of a model:
 - Rather than how fast it takes to classify or build models, scalability, etc.
- Confusion Matrix:
 - Let's we elaborate with a case example.

Case example of application Confusion Matrix

Chest Pain	Good Blood Circ.	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
...	...	...	...	...



Imagine that we have this medical data...

Case example of application Confusion Matrix

We've got some clinical measurements...



Chest Pain	Good Blood Circ.	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
...	...	...	...	...

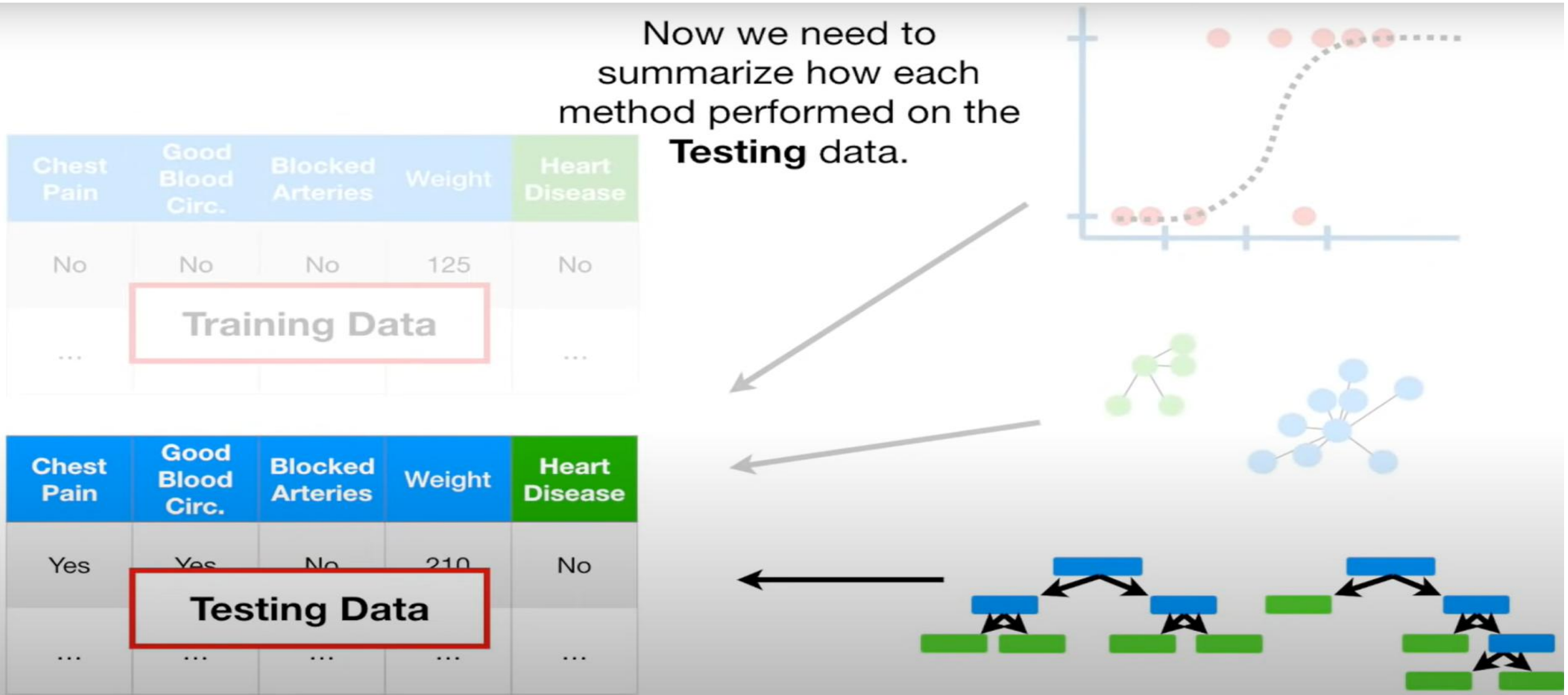
Case example of application Confusion Matrix

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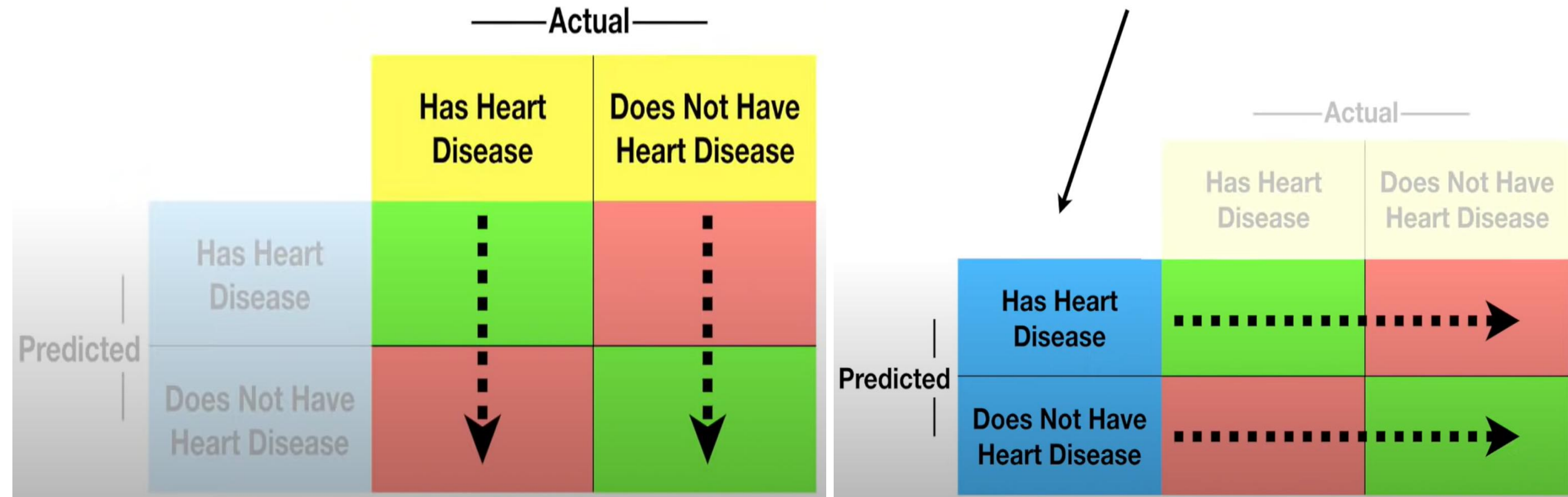
...and we want to apply a machine learning method to them to predict whether or not someone will develop heart disease.



Case example of application Confusion Matrix



Case example: Actual and Predicted data



Most widely-used metric of Accuracy is:
$$= \frac{TP + TN}{TP + TN + FP + FN}$$

Case example of application Confusion Matrix

		Actual	
		Has Heart Disease	Does Not Have Heart Disease
Predicted	Has Heart Disease		
	Does Not Have Heart Disease		

Case example: True Positive

These are the patients that *had heart disease* that were correctly identified by the algorithm.

		Actual	
		Has Heart Disease	Does Not Have Heart Disease
Predicted	Has Heart Disease	True Positives	
	Does Not Have Heart Disease		

Case example: True Negative

The **True Negatives** are in the bottom right-hand corner.

These are the patients that did not have heart disease and were correctly identified by the algorithm as not having heart disease.

		Actual	
		Has Heart Disease	Does Not Have Heart Disease
Predicted	Has Heart Disease	True Positives	
	Does Not Have Heart Disease		True Negatives

Case example: False Negative

False Negatives are when a patient has heart disease, but the algorithm said they didn't.

		Actual	
		Has Heart Disease	Does Not Have Heart Disease
Predicted	Has Heart Disease	True Positives	
	Does Not Have Heart Disease	False Negatives	True Negatives

Case example: False Positive

False Positives are patients that do not have heart disease, but the algorithm says they do.

		Actual	
		Has Heart Disease	Does Not Have Heart Disease
Predicted	Has Heart Disease	True Positives	False Positives
	Does Not Have Heart Disease	False Negatives	True Negatives

Case example: Record evaluation [TP]

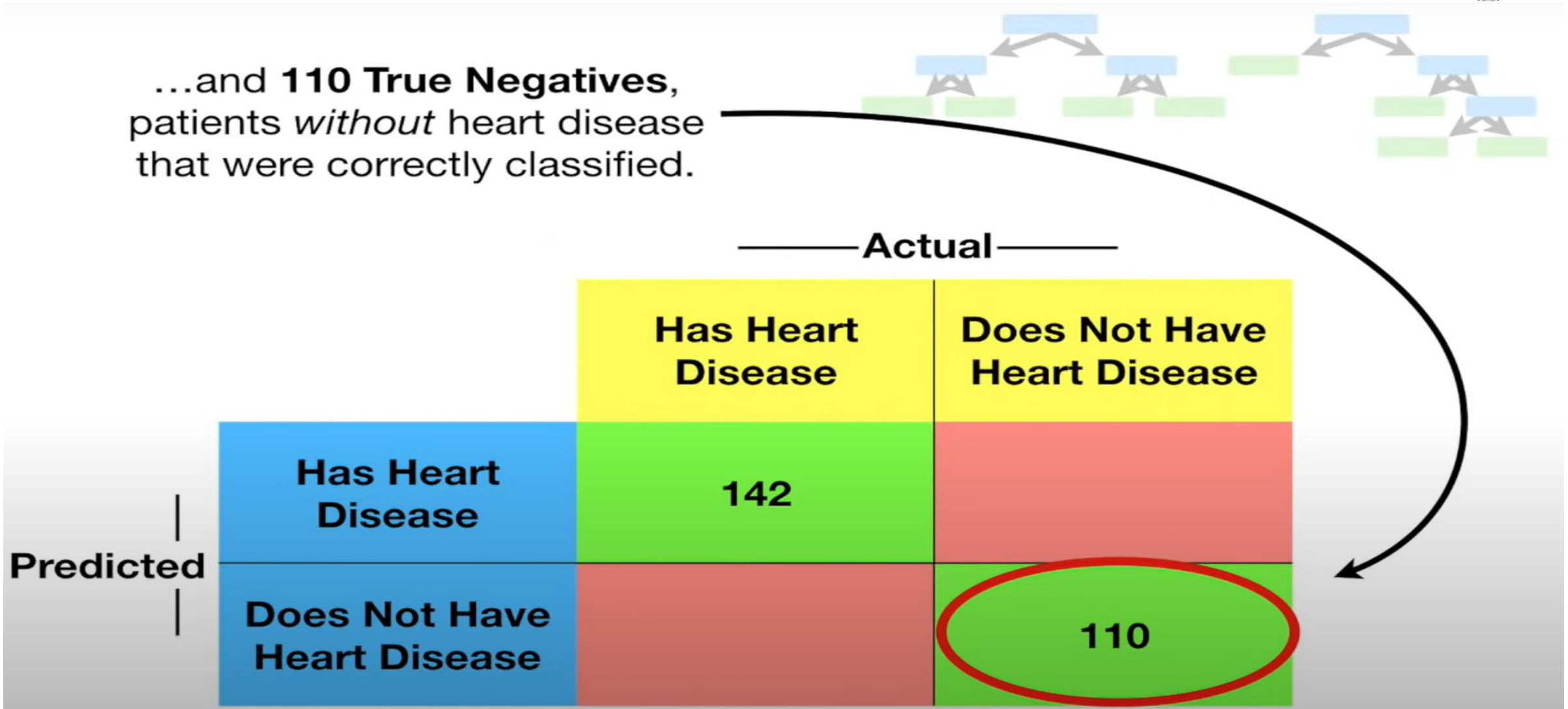
There were **142 True Positives**, patients *with* heart disease that were correctly classified...



		Actual	
		Has Heart Disease	Does Not Have Heart Disease
Predicted	Has Heart Disease	142	
	Does Not Have Heart Disease		

Case example: Record evaluation [TN]

...and **110 True Negatives**,
patients *without* heart disease
that were correctly classified.



Case example: Record evaluation [FN]

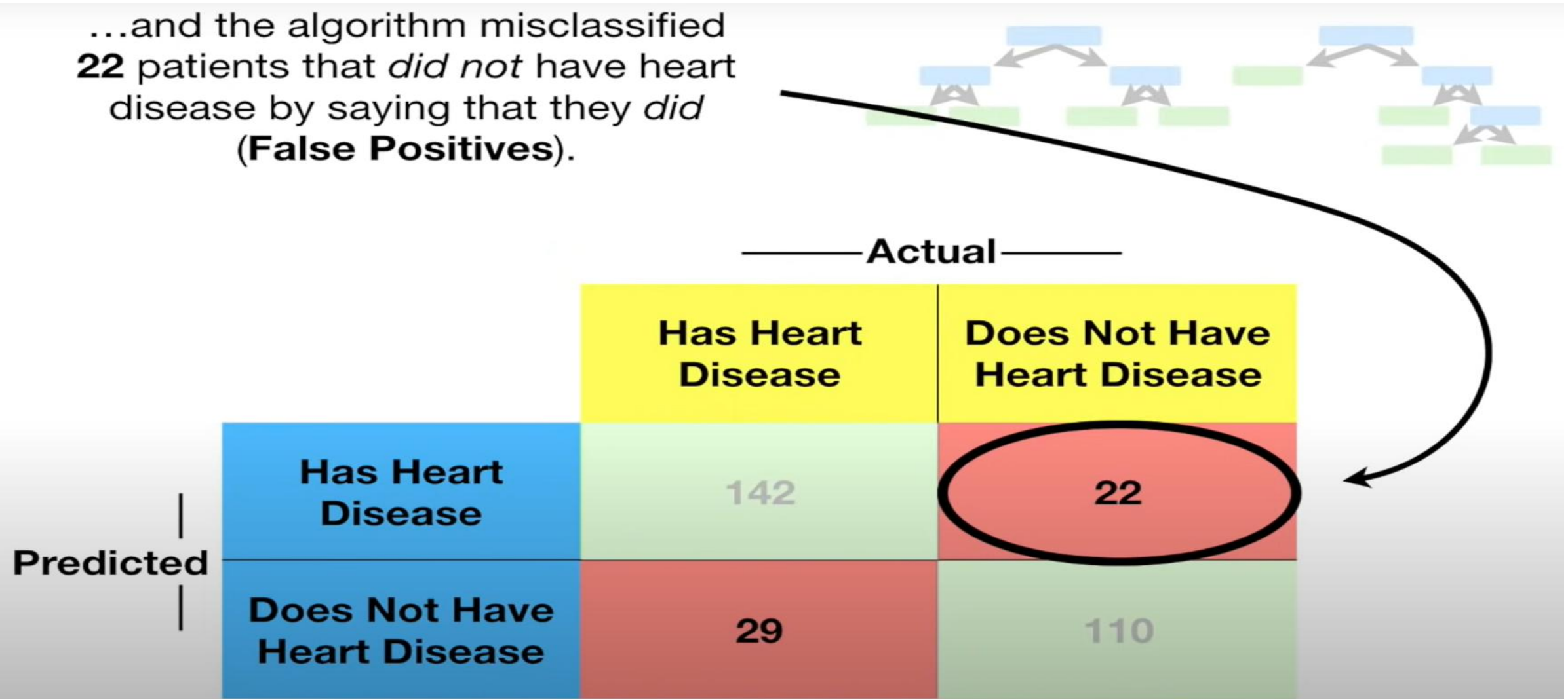
However, the algorithm misclassified **29** patients that *did* have heart disease by saying that they *did not* (**False Negatives**)...



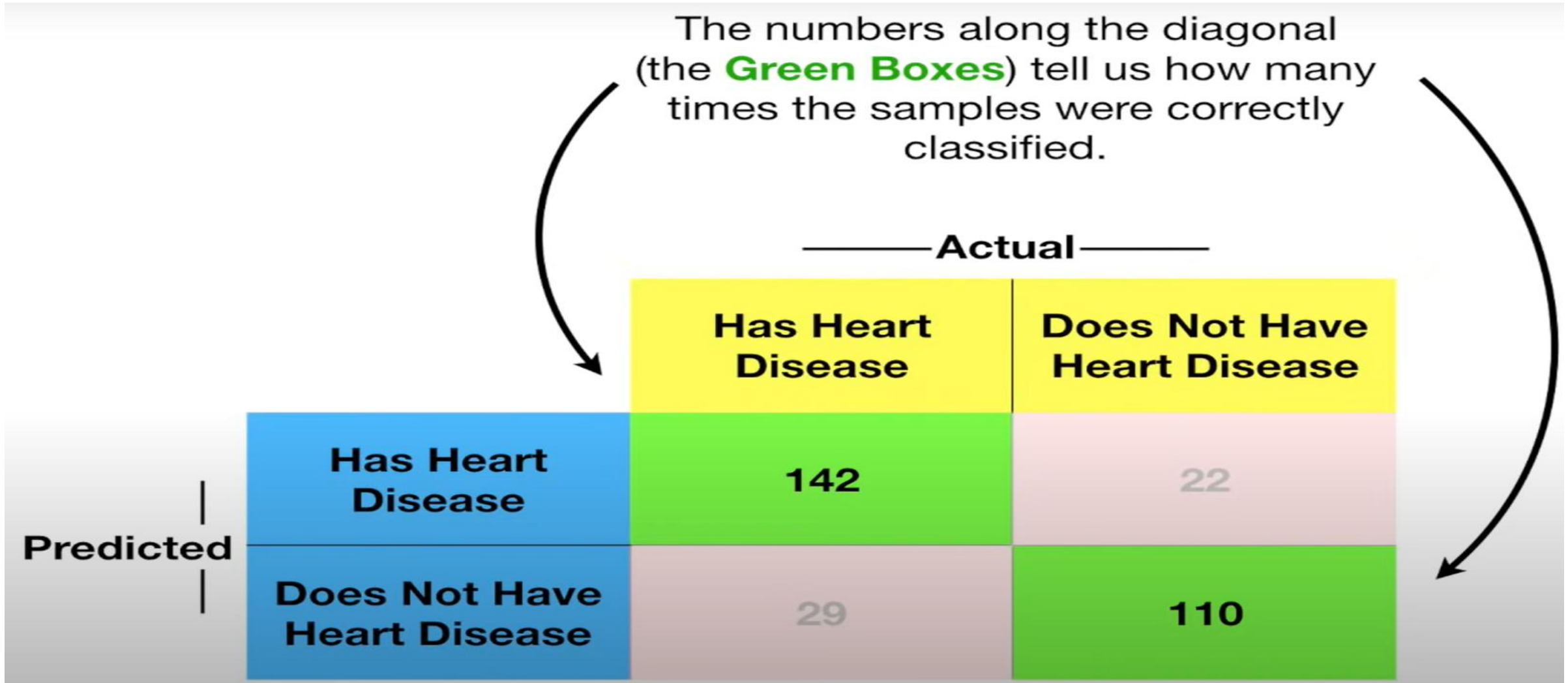
		Actual	
		Has Heart Disease	Does Not Have Heart Disease
Predicted	Has Heart Disease	142	
	Does Not Have Heart Disease	29	110

Case example: Record evaluation [FP]

...and the algorithm misclassified **22** patients that *did not* have heart disease by saying that they *did* (**False Positives**).



Case example: Green Boxes



Case example: Red Boxes



The numbers *not* on the diagonal (the **Red Boxes**) are samples the algorithm messed up.



		Actual	
		Has Heart Disease	Does Not Have Heart Disease
Predicted	Has Heart Disease	142	22
	Does Not Have Heart Disease	29	110

Limitation of Accuracy

- Consider a 2-class problem:
 - Number of Class 0 examples = 9990;
 - Number of Class 1 examples = 10.
- If model predicts everything to be class 0, accuracy is $9990/10000 = 99.9\%$:
 - Accuracy is misleading because model does not detect any class 1 example.

Precision vs. Recall vs. F-Measure

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision: Precision is the ratio of correctly predicted positive observations to the total predicted positives. It is a measure of how accurate the positive predictions are.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall (Sensitivity or True Positive Rate): Recall is the ratio of correctly predicted positive observations to all the actual positives. It is a measure of how well the model captures all the positive instances.

$$\text{F - measure} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F-measure is then computed as the harmonic mean of precision and recall

ROC (Receiver Operating Characteristic)

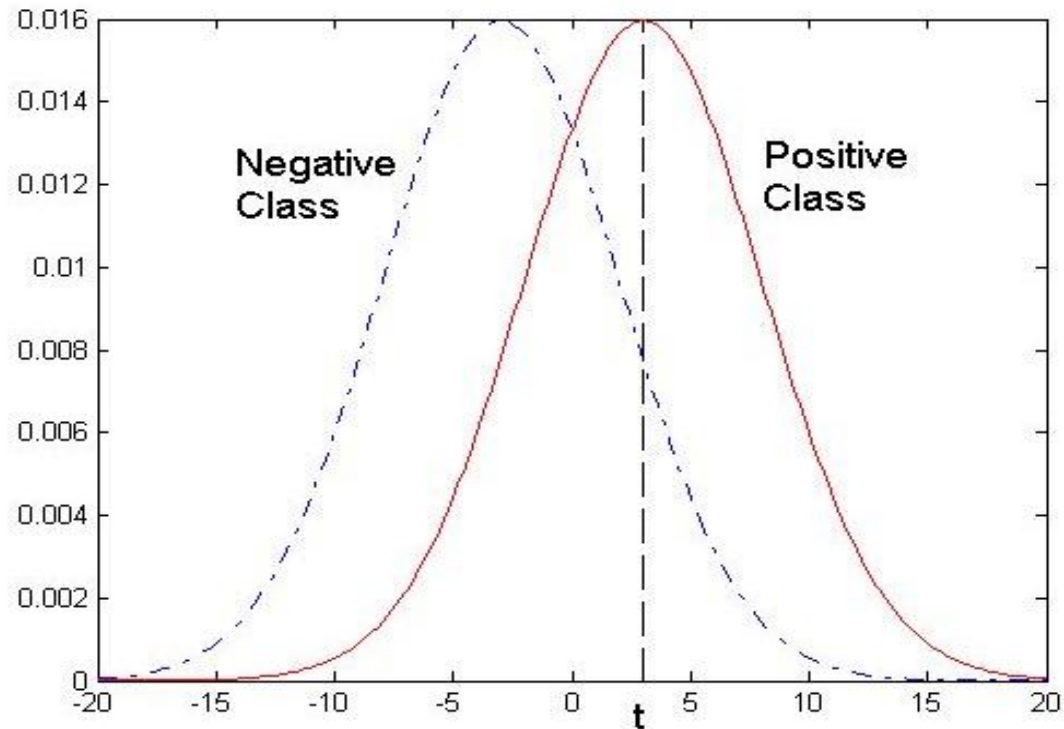
- Developed in 1950s for signal detection theory to analyze noisy signals:
 - Characterize the trade-off between positive hits and false alarms.

- ROC curve plots TP (on the y-axis) against FP (on the x-axis);

- Performance of each classifier represented as a point on the ROC curve:
 - Changing the threshold of algorithm, sample distribution or cost matrix changes the location of the point.

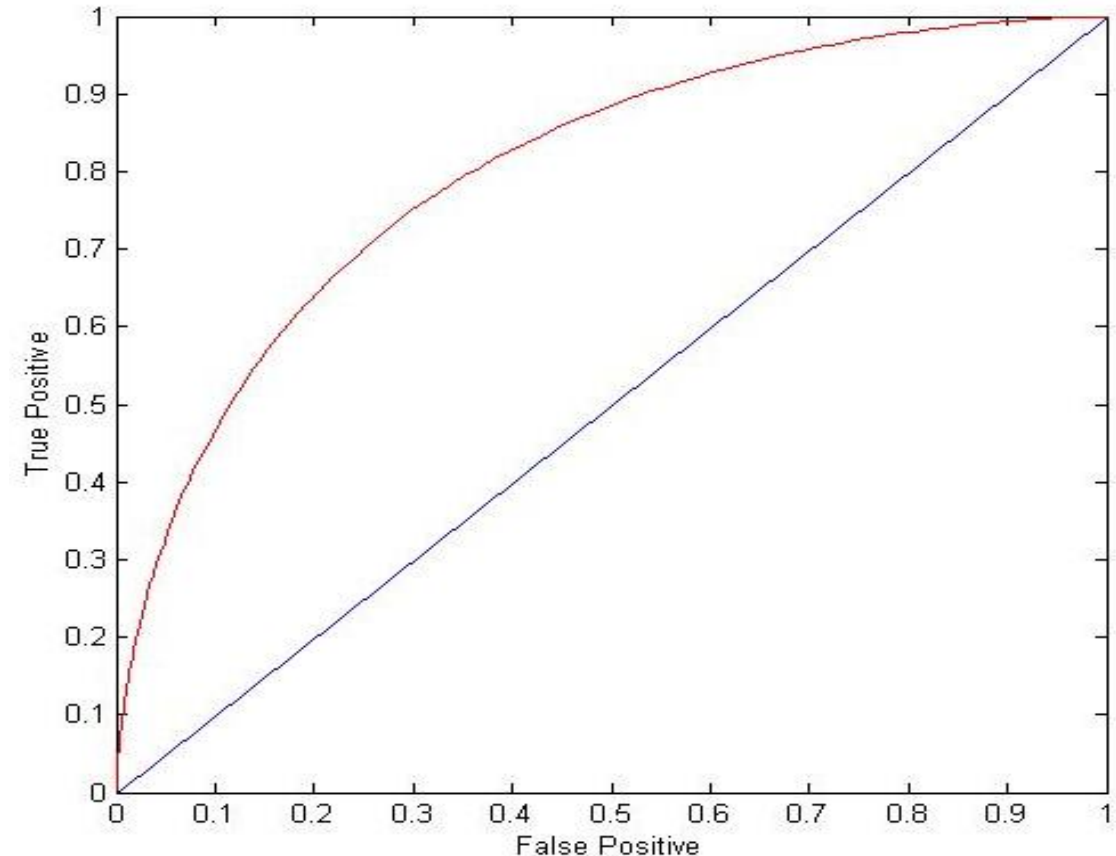
ROC Curve

- 1-dimensional data set containing 2 classes (positive and negative);
- any points located at $x > t$ is classified as positive.



At threshold t :

TP=0.5, FN=0.5, FP=0.12, FN=0.88



ROC Curve

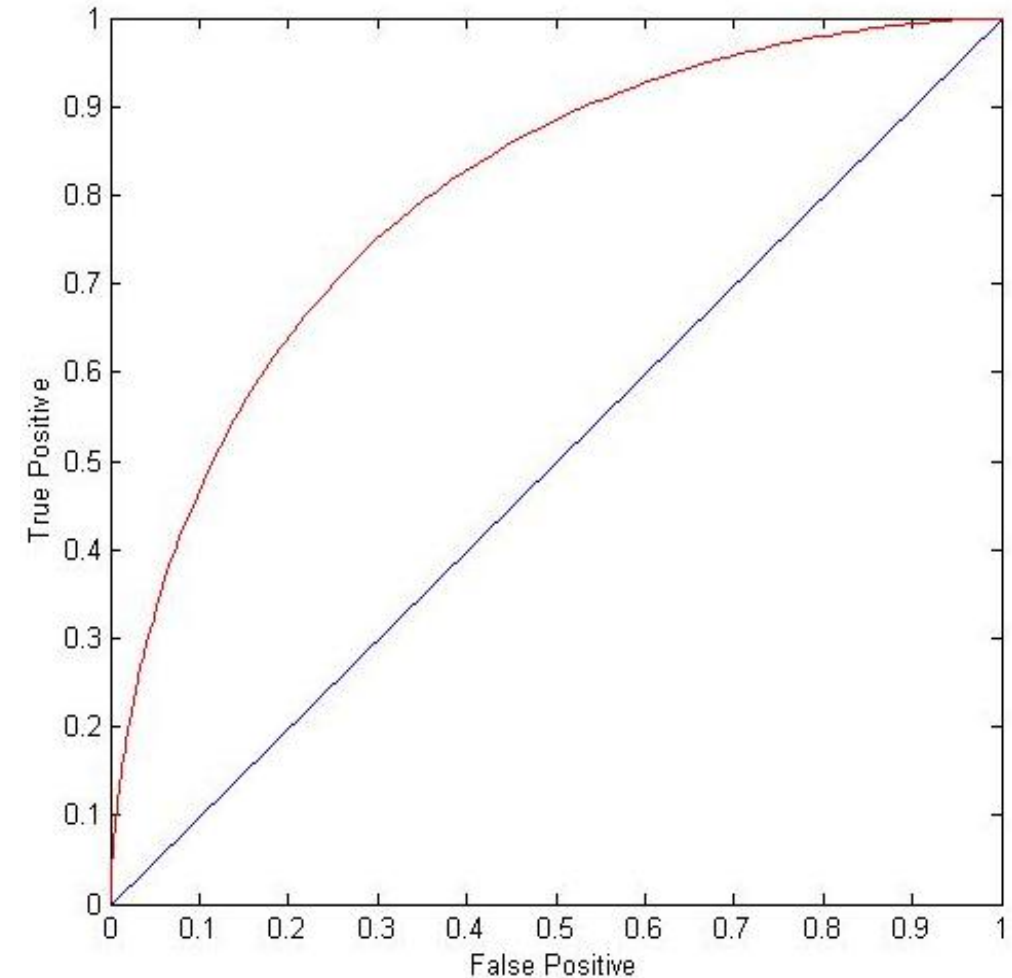


➤ (TP,FP):

- (0,0): declare everything: *to be negative class*;
- (1,1): declare everything: *to be positive class*;
- (1,0): *ideal*;

➤ Diagonal line:

- Random guessing;
- Below diagonal line:
 - prediction is opposite of the true class



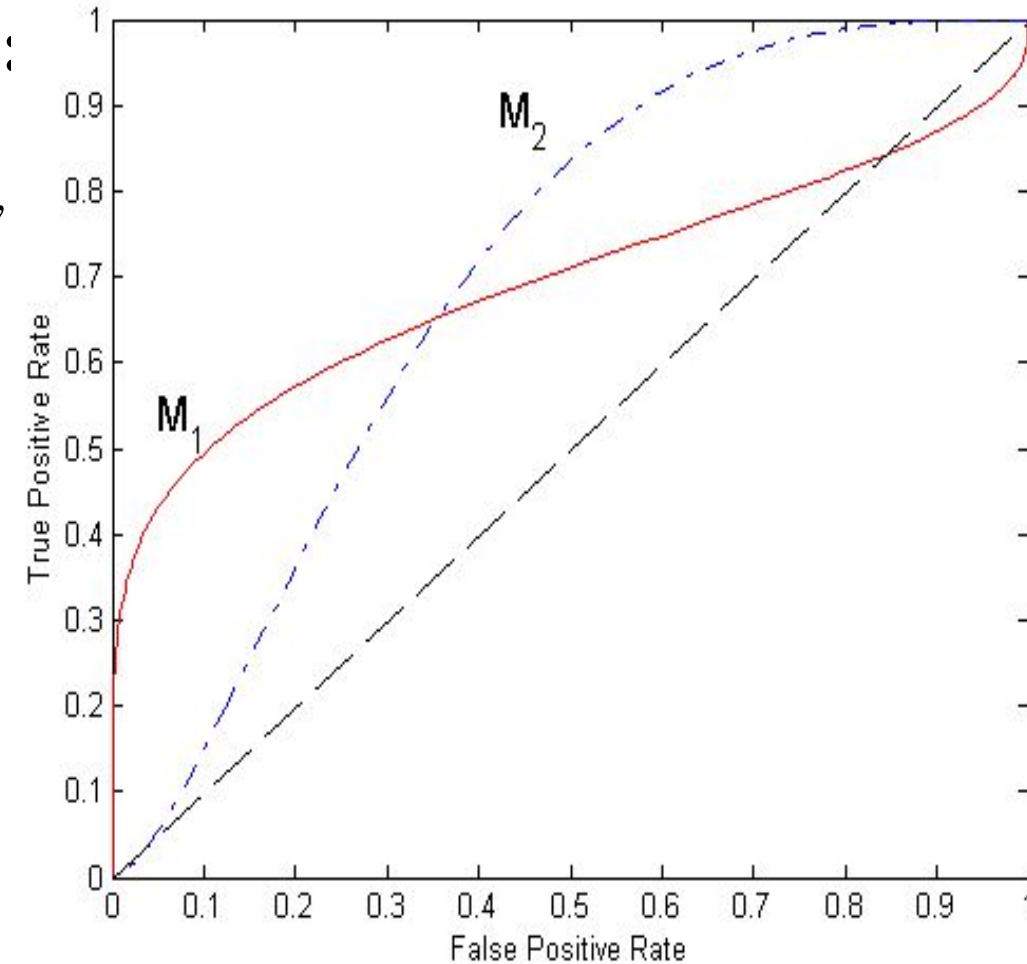
Using ROC for Model Comparison

➤ No model consistently outperform the other:

- **M₁ is better for small FPRate:**
 - *A small FPR is crucial because you want to avoid falsely diagnosing a healthy person with a disease.*
- **M₂ is better for large FPRate:**
 - A higher FPR might be tolerable since marking a legitimate email as spam is less critical.

➤ Area Under the ROC curve:

- **Ideal:**
 - Area = 1
- **Random guess:**
 - Area = 0.5



Methods of Estimation

➤ Holdout:

- *Reserve 2/3 for training and 1/3 for testing;*

➤ Stratified sampling:

- *Oversampling vs under-sampling;*

➤ Cross validation:

- *Partition data into k disjoint subsets;*
- *k -fold: train on $k-1$ partitions, test on the remaining one;*
- *Leave-one-out: $k=n$.*

10-Fold Cross Validation (Example)

➤ What if we don't have enough data to set aside a test dataset?

➤ Cross-Validation:

- Each data point is used both as train and test data;

➤ Basic idea:

- Fit model on 90% of the data and test on other 10%;
- Now, do this on a different 90/10 split;
- Cycle through all 10 cases;
- 10 “folds” a common rule of thumb.

10-Fold Cross Validation (Example)

- Divide data into 10 equal pieces $P_1 \dots P_{10}$.
- Fit 10 models, each on 90% of the data;
- Each data point is treated as an out-of-sample data point by exactly one of the models.

model	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
1	train	train	train	train	train	train	train	train	train	test
2	train	train	train	train	train	train	train	train	test	train
3	train	train	train	train	train	train	train	test	train	train
4	train	train	train	train	train	train	test	train	train	train
5	train	train	train	train	train	test	train	train	train	train
6	train	train	train	train	test	train	train	train	train	train
7	train	train	train	test	train	train	train	train	train	train
8	train	train	test	train	train	train	train	train	train	train
9	train	test	train	train	train	train	train	train	train	train
10	test	train	train	train	train	train	train	train	train	train

10-Fold Cross Validation (Example)

➤ Collect the scores from the red diagonal...

model	P1	P2	P3	P4	P5	P6	P7	P8	P9
1	train	train	train	train	train	train	train	train	train
2	train	train	train	train	train	train	train	train	test
3	train	train	train	train	train	train	train	test	train
4	train	train	train	train	train	train	test	train	train
5	train	train	train	train	train	test	train	train	train
6	train	train	train	train	test	train	train	train	train
7	train	train	train	test	train	train	train	train	train
8	train	train	test	train	train	train	train	train	train
9	train	test	train	train	train	train	train	train	train
10	test	train	train	train	train	train	train	train	train

Questions?

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